

University of Illinois Chicago | College of Engineering

Quantum Research

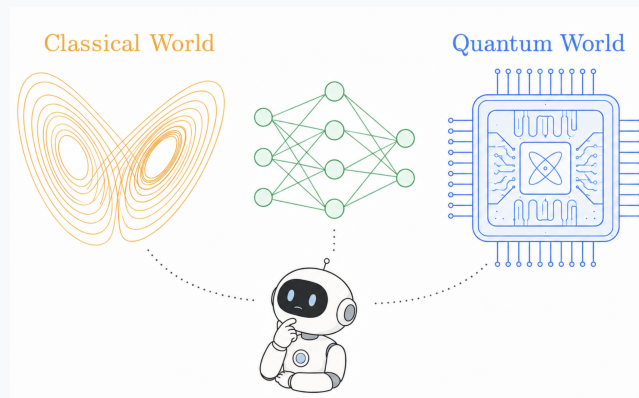
Mondragon-Shem Quantum Group

Student: Marcos Sandoval

AI Design of Quantum Processors

🎯 Research Project Goal

Overarching Goal: Predict properties of quantum processors using classical data. The project proceeds through three research components:



⚙️ Component 1

Classical Data Generation

Simulate classical mechanical systems using Newton's equations of motion.

⚙️ Component 2

Quantum Data Generation

Calculate quantum properties analytically and with QuTiP.

🧠 Component 3

ML Training

Choose, train, and validate an ML algorithm using the generated data.

📖 References & Resources

- **PI:** Feel free to reach out with any questions you may have.
- **QuTiP** (quantum toolbox in Python): qutip.org
- **Quantum Mechanics** (Essler, Oxford): [lecture notes PDF](#)
- **Superconducting quantum processors** (basic overview): [PennyLane tutorial](#)

★ Best Practices

🔗 Coding

- **Write readable code.** Use descriptive variable names, break work into short functions, and add a one-line comment for every non-obvious step. All code should be written in **Jupyter notebooks**.
- **Use a shared GitHub repository.** All your notebooks and scripts must live in the group GitHub repo. Push your changes regularly and keep the PI added as a collaborator so progress is always visible.

📊 Plots

- **Label everything.** Every axis must carry a label with units (or state dimensionless). Every curve needs a legend entry. Every figure needs a caption that explains what is shown and what the key takeaway is.
- **Choose colors intentionally.** Use perceptually uniform colormaps (e.g. `viridis`, `plasma`) so that differences in data are faithfully represented on screen and in print.
- **Size fonts for the medium.** Set `fontsize` so that tick labels, axis titles, and legends remain legible once the figure is scaled to a slide or a printed page.

👤 Meeting Slides

- **Organize your latest calculations into slides.** For each meeting, collect your most recent results as polished plots in a slide deck. Annotate the key features directly on the figures rather than describing them verbally.
- **One idea per slide.** State the main takeaway in the slide title as a complete sentence so the audience understands your point even if they only skim the titles.
- **Structure each update in three steps:**
 1. *Context:* discuss why you were doing this calculation and what question it was meant to answer.
 2. *Results:* present what you found, supported by figures.
 3. *Open questions:* describe what you do not yet understand and what you would like to do or discuss next.

Research Component 1: Classical Data Generation

Task 1: Classical Harmonic Oscillator: Energy and Phase Space

Coding

The harmonic oscillator is one of the most important models in physics. It describes small oscillations near a stable equilibrium and appears in mechanics, circuits, vibrations, and quantum hardware. In this task you will study the classical energy function and its phase-space structure.

- (a) State the classical energy $E(x, p)$ of a one-dimensional harmonic oscillator in terms of position x , momentum p , mass m , and angular frequency ω .
- (b) The equations of motion can be derived from the energy by treating it as the Hamiltonian $H(x, p)$. Starting from Hamilton's equations,

$$\dot{x} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial x},$$

derive the classical equations of motion for $x(t)$ and $p(t)$. Combine them to obtain the familiar second-order equation for $x(t)$.

- (c) Make a plot of the classical energy $E(x, p)$ in phase space. Use either a contour plot or a density plot over a two-dimensional grid of (x, p) values. Explain the shape of the constant-energy curves.

Task 2: Classical Harmonic Oscillator: Dynamics and Trajectories

Coding

The energy function tells us which phase-space curves are allowed; dynamics tells us how the system moves along those curves. In this task you will simulate and visualize classical harmonic oscillator trajectories.

- (a) Solve the classical equations of motion derived in Task 1. Choose one initial condition (x_0, p_0) and make a phase-space plot of the trajectory $(x(t), p(t))$. Indicate the initial point on the plot.
- (b) Repeat the simulation for a random set of initial conditions. Plot all trajectories on the same axes and explain how the trajectory shapes depend on the initial energy.

Research Component 2: Quantum Data Generation

Task 1: Quantum Harmonic Oscillator: Operators and Energy Spectrum

Conceptual Coding

In quantum mechanics, physical observables are represented by operators and the energy spectrum is discrete. In this task you will construct the quantum Hamiltonian and compute its eigenvalues.

- (a) In quantum mechanics, position and momentum become the operators $\hat{x}$ and $\hat{p}$. Using QuTiP, construct the truncated matrix representations of

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger), \quad \hat{p} = -i\sqrt{\frac{\hbar m\omega}{2}} (\hat{a} - \hat{a}^\dagger),$$

and

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2.$$

Make `imshow` plots of the absolute values of the matrix elements of $\hat{x}$, $\hat{p}$, and $\hat{H}$.

- (b) The possible energies of a quantum system are the eigenvalues of the Hamiltonian. Use QuTiP to calculate the eigenvalues of $\hat{H}$. Plot the first several eigenvalues versus the level index n , and compare your numerical values to the analytic expression

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

- (c) Briefly explain what changed, and what stayed the same, when moving from the classical energy function $E(x, p)$ to the quantum Hamiltonian matrix $\hat{H}$.

Task 2: Quantum Harmonic Oscillator: State Dynamics and Phase Space

Conceptual Coding

Quantum dynamics are governed by the Schrödinger equation, and quantum states can be visualized in phase space using the Wigner function. In this task you will simulate quantum time evolution and compare it to classical motion.

- (a) Use QuTiP's `sesolve` function to solve the Schrödinger equation,

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle,$$

for the following initial states:

- i. an energy eigenstate $|n\rangle$, such as $|0\rangle$ or $|1\rangle$,

- ii. the normalized superposition $(|0\rangle + |1\rangle)/\sqrt{2}$,
- iii. a displaced Gaussian state, implemented as a coherent state $|\alpha\rangle$ or as $\hat{D}(\alpha)|0\rangle$.

Store the quantum state at each time step so that you can visualize its dynamics.

- (b) The Wigner function $W(x, p)$ is a phase-space representation of a quantum state. It allows us to compare quantum states to classical motion in the same (x, p) plane. For each initial state in part (a), use **QuTiP** to compute the Wigner function over a grid of x and p values. Make a static plot of $W(x, p)$ at $t = 0$.
- (c) Make a movie of the Wigner function dynamics for each of the three initial states from part (a). Use a fixed color scale for each movie so that changes in the shape and sign of $W(x, p)$ are visually meaningful.
- (d) For each quantum trajectory, calculate the expectation values $\langle \hat{x} \rangle(t)$ and $\langle \hat{p} \rangle(t)$. Plot $\langle \hat{p} \rangle$ versus $\langle \hat{x} \rangle$ in phase space. On the same axes, overlay the classical solution with initial condition

$$x_0 = \langle \hat{x} \rangle(0), \quad p_0 = \langle \hat{p} \rangle(0).$$

Discuss when the expectation-value motion agrees with the classical trajectory and what information is lost when only the averages are plotted.

- (e) Write a short reflection, no more than one page, explaining what each representation tells you: energy contours, matrix elements, eigenvalues, Wigner functions, and expectation-value phase-space plots.

Research Component 3: ML Training

 Tasks for Research Component 3 will be introduced after completing Components 1 and 2.